STARFORGE

A pocket atlas of stellar engines

A star is a machine that burns time into light.

Eight chambers. One page each. Walk them in order.

How to walk a chamber

This atlas is not a story you finish. It is a hall of rooms. Each page is one engine, cut open and lit from the inside. You do not skim. You stand in the room until the machine makes sense, then you step through the wall into the next.

Four instruments hang on every wall: mass, which sets the appetite of the fire; light, which is the waste heat of that appetite; color, which is a temperature you can see; and time, which is how long the fuel can last. Read them as a set. A number alone is a souvenir. Four numbers are a diagnosis.

The four instruments

MASS	How much matter the engine can crush. Unit: solar masses.
LIGHT	Power poured into space. Unit: solar luminosities.
COLOR	Surface temperature, written as spectral class O through M.
TIME	Main-sequence lifetime. Heavy stars spend their fortune fast.

The mass–luminosity ledger

Double a star's mass and you do not double its light. You multiply it. The engine is nonlinear: core pressure climbs, fusion rate explodes, and the surface becomes a furnace that cannot be sustained for long. This table is the whole argument in six rows.

Class	Mass (☉)	Luminosity (☉)	Lifetime (yr)	Temp (K)
O5	40	400000	1,000,000	40000
B0	17	20000	10,000,000	25000
A0	2.7	55	400,000,000	9500
G2 (Sun)	1.0	1	10,000,000,000	5772
K5	0.67	0.17	30,000,000,000	4400
M0	0.51	0.08	70,000,000,000	3800

Read the last two columns against each other. Forty thousand kelvin costs a million years. Thirty-eight hundred kelvin buys seventy billion. The sky is full of thrifty red engines because the spendthrifts have already left.

The boiling core

Fifteen million kelvin is not a metaphor. It is the temperature at which hydrogen nuclei move fast enough to tunnel through their own repulsion and become helium. In the Sun this happens in a core about 0.2 solar radii across, under a density of 150 grams per cubic centimeter — more like white-hot metal than gas.

Energy does not beam out. Photons are born as gamma rays and then play a three-dimensional drunkard's walk through the radiative zone, colliding for about 100,000 years before they reach the convective envelope. What you see as sunlight is a delayed letter from the deep past of the core.

Sun-as-engine, present tense

Core temperature	15,000,000 K
Core density	150 g/cm³
Photon random-walk	100,000 years
Neutrino escape	2.3 seconds
Hydrogen fused / sec	620,000,000 tons
Mass lost as light / sec	4,300,000 tons

Neutrinos skip the drunkard's walk. They leave the core in 2.3 seconds and pass through the Earth at night as if the planet were a rumor. A star is therefore two clocks at once: a slow clock of light, and a fast clock of ghosts.

Spectral engines

Annie Jump Cannon sorted the sky into a temperature ladder: O B A F G K M. The mnemonic is a joke. The science is a thermometer you can hang on a spectrum. Blue-white is hungry. Red is patient. The Sun is a G2, which is to say a yellow-white moderate, neither miser nor spendthrift.

Class	Color	Temp (K)	Example	Fate
O	blue-white	30000–50000	Alnitak	core collapse
B	blue	10000–30000	Rigel	core collapse
A	white	7500–10000	Sirius	white dwarf
F	yellow-white	6000–7500	Procyon	white dwarf
G	yellow	5200–6000	the Sun	white dwarf
K	orange	3700–5200	Arcturus	white dwarf
M	red	2400–3700	Proxima	long fade

Only O and B stars die as collapses that can mint neutron stars or black holes. Everything cooler than about 8 solar masses will one day be a white dwarf: an Earth-sized ember of degenerate carbon, cooling for longer than the universe has so far lasted.

A census of neutrinos

Every second, about 65 billion solar neutrinos pass through each square centimeter of your body. They are not a weather. They are a correspondence. Most were written 8.3 minutes ago in the proton–proton chain; a few are fossils from supernovae and the big bang.

Solar neutrinos through a fingernail / sec	65,000,000,000
Time to cross the Sun from core	2.3 seconds
Time for light to do the same	100,000 years
Earth as neutrino shield	effectively transparent
Supernova 1987A neutrinos detected	25
Rest mass	< 0.8 eV

In 1987, twenty-five neutrinos arrived from a star that died in the Large Magellanic Cloud 160,000 years earlier. They came three hours before the light. That is the holodeck trick the universe already performs: the ghost of an engine reaches you before the picture does.

The giant's last ledger

When core hydrogen is gone, the engine does not politely stop. The core contracts, shell burning ignites, and the envelope inflates into a red giant. For a Sun-like star the sequence is a timetable, not a mystery.

Stage	Year offset	Radius (☉)	Luminosity (☉)	Note
main sequence now	0	1	1	core H fusion
subgiant	+5.4e9	2.3	2.2	H shell ignites
red giant branch	+7.1e9	150	2000	core helium ash
helium flash	+7.7e9	10	50	core He ignition
horizontal branch	+7.8e9	10	50	quiet He burning
AGB pulse	+8.0e9	200	3000	shells double
planetary nebula	+8.1e9	—	—	envelope ejected
white dwarf	+8.2e9	0.01	0.001	Earth-sized ember

The red giant the Sun becomes will swallow Mercury and Venus and may lick the Earth. Then it will shrug off its atmosphere as a planetary nebula — a poor name for a rich event — and leave a white dwarf of about 0.54 solar masses. That ember is the original engine, retired, still hot, no longer employed.

The sky is a machine

Walk back through the chambers and the night stops being a decoration. Every naked-eye star is an engine whose ledger you can now roughly guess: color is temperature, brightness plus distance is power, power plus mass is a lifetime. The constellations are not stories. They are a shop floor.

You do not need a telescope to begin. You need the habit of treating a page as a room. Stand in it. Let the numbers become objects. When the next page opens, the previous room goes dark behind you, the way a holodeck does, and the new engine takes the floor.

Close the book. Look up. The machines are still on.

End of volume I · drop another book to raise a new hall